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SEA SAFE TRACE SYSTEM

Oil spill detection system

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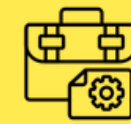
WHAT IS THE PROBLEM?

- **Oil spills silently happening in the oceans and seas**
- **Polluted water marine life and coastal communities**
- **It is Difficult to know which ship caused the spill**
- **Many spills go unreported or unnoticed**



SOLUTION?

- **Combine satellite data + ship tracking data**
- **Automatically detect oil spills in near real time**
- **Track back and identify the responsible ship**
- **Supports authorities with clear visual evidence**



HOW IT WORKS?

- **Satellite detect oil slicks on the sea surface**
- **The system checks time and location of the spill**
- **Ship GPS/AIS data is matched with the spill area**
- **The most likely ship is identified using smart filtering**



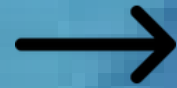
WHY IT IS NEEDED?

- **Protect oceans , fish , and marine ecosystems**
- **Hold Polluters accountable**
- **Reduce illegal oil dumping**
- **Enable faster response and cleanup**
- **Support environmental laws and safety**

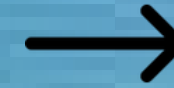
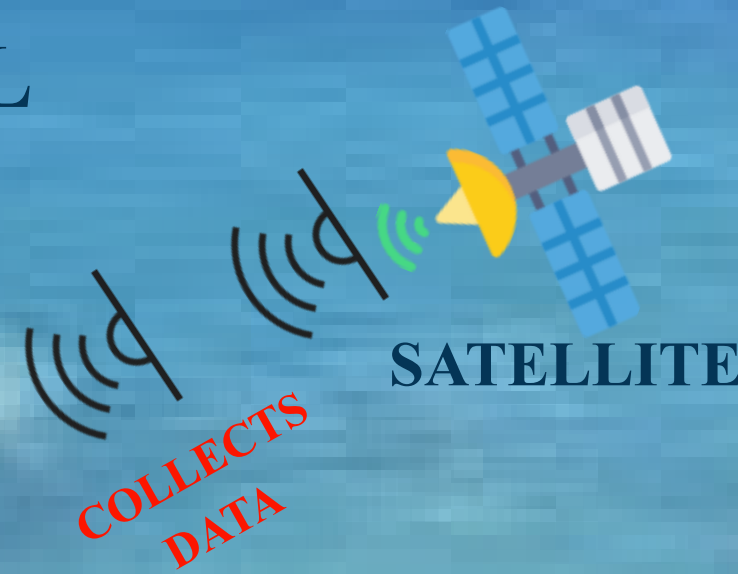


WORK FLOW

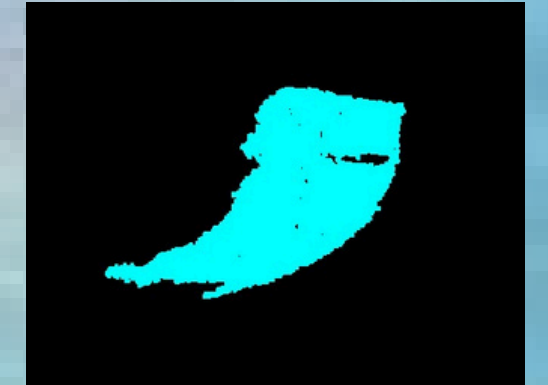
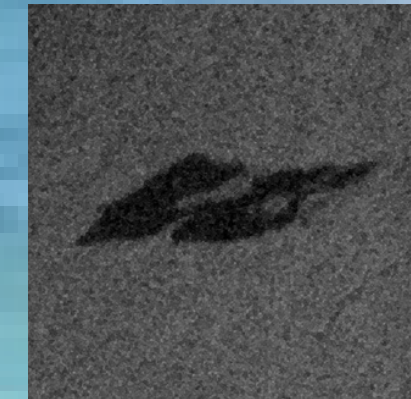
SHIP



SHIP SPILLING OIL



SAR IMAGE DETECTION



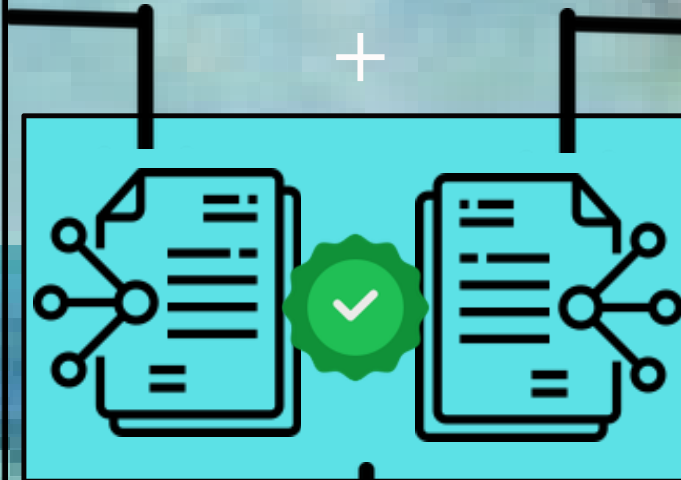
GPS TRACKING



COLLECTS DATA



AIS INFORMATION
THROUGH GPS



+

! ALERT TO SHIP PORT

